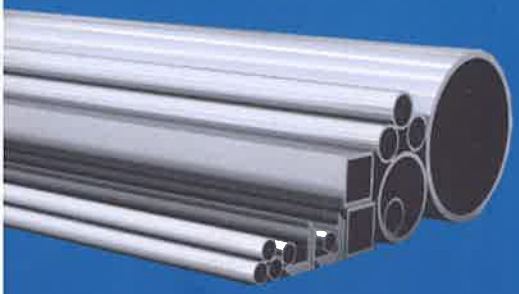
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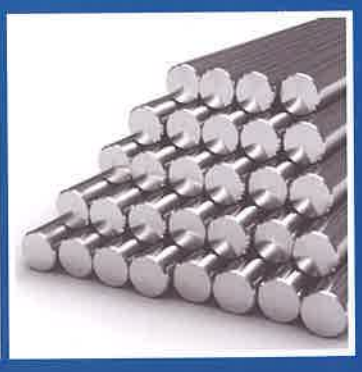
STEEL INDIA



MFRS., STOCKISTS & SUPPLIERS OF :

- PIPES & TUBES
- SHEET, PLATE, COILS & RODS
- FLANGES
- FASTENERS
- BUTTWELD FITTINGS
- FORGED FITTINGS
- VALVES



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Steel India Co.
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Company Profile

Steel India Co. Chennai Pvt Ltd. was commented in the year 1990. Steel India Co. Chennai Pvt Ltd. is one of the trustworthy manufacturer, importer, dealer, supplier and stockist of Stainless Steel products. Import these products

We never compromise on quality, and this makes us a preferred choice among our competitors. The best part is that quality norms are not only followed in procuring, the broad spectrum of this stretches to delivery as well. After receiving the request from the concerned clients, we deliver these products within the defined time frame. We keep constant touch with the clients to know their exact requirements.

Our mentor has been the guiding light since decades. Owing to his industrial experience and managerial skills, we are able to attain huge success rate. By adopting fair business policies and transparency, we are able to attain complete client satisfaction.

Product Portfolios

We are instrumental in offering wide assortment of stainless steel products for the clients. Since the day of commencement, we are executing our business operations smoothly our products plethora includes

● Stainless Steel ● Carbon Steel ● Mild Steel ● Plates ● Pipes ● Tubes ● Flats ● Angles ● Rods ● Channel ● Sheets ● Coils ● Chequered

Client Satisfaction

Being the client oriented organization; we are able to cater qualitative products for the clients. Keeping in mind the convenience of our clients, we provide products in diverse specifications in order to fulfill their varied requirements. Furthermore, due to quick deliveries of the orders, our competent professionals have supported us in earning great respect across the region. Your valuable feedback is widely appreciated as if helps in upgrading our services.

Industrial experience blended with quality, commitment, promptness
punctuality & trust worthiness has been our
"SECRET OF SUCCESS"
and the quality oriented customer once experienced our empathetic
approach really wants us by all means which has been our
"STRENGTH"
we enjoy serving you
The Owners

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Steel India Co.
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Product Range - Stainless Steel

Stainless Steel is available in a veriety of forms. We have a ready stock of the following product range :

S.No.	Product Description	Thickness (mm)	Width (mm)	Length (mm)
1.	S.S. Foils	0.05 - 0.50	50-600	Any Length
2.	S.S. Coils	0.20 - 12.0	600-2500	Any Length
3.	S.S. Sheets	0.40 - 4.0	900-2000	Any Length
4.	S.S.Plates	4.00 - 100.0	1000-2500	Any Length
5.	S.S. Strips	0.20 - 3.0	9-600	Any Length
6.	S.S. Flats	3.0 - 50.0	Any width	Any Length
7.	S.S. Pipes	0.5-25.0 (Wall)	3.0-610.0 (O.D)	Any Length
8.	S.S. Wires	0.05 - 2.0 (Ø)	-	Any Length
9.	S.S. Rods	2.0 - 400.0 (Ø)	-	Any Length
10.	S.S. Rounds	0.10 - 50.0 (Ø)	10-2000 (Ø)	-
11.	S.S. Flanges	Any Size	Any Size	Any Size
12	S.S. Channels	Any Size	Any Size	Any Size
13.	S.S. Angles	Any Size	Any Size	Any Size
14.	S.S. Pipe Fittings	Any Size	Any Size	Any Size
15.	S.S. Balls	Any Size	Any Size	Any Size
16.	S.S. Rings	Any Size	Any Size	Any Size

Any Non Standard Size Can Be Provided

Surface Finishes - Stainless Steel

Surface finish is an important element in any specification for stainless steel. SSTC offers you a wide range of finishes with expert guidance on which finish would be best suited for your application.

Hot Rolled

No.1 Finish : Slabs are hot rolled to plate / coils, annealed shot blasted and pickled. This results in a dull, slightly rough surface; quite suitable for industrial application.

Cold Rolled

2D Finish : Material with No. 1 finish is cold rolled, annealed and pickled, This results in a dull but superior finish, when compared to No. 1 finish. it is suitable for severe deep drawing as the dull surface retains the lubricant during the drawing operation.

2B Finish : Material with 2D finish is given a subsequent light skin pass operation between polished rolls. It is brighter than 2D and is semi - reflective.

No3 Finish : This is ground unidirectional uniform finish obtained with 100 - 120 grit abrasive. It is a good intermediate finish for surfaces which would require finer finish after the fabrication / forming process.

No4 Finish : This is ground unidirectional uniform finish obtained with 120 - 150 grit abrasive. It is not highly reflective but is suited for components which would suffer from rough handling.

BA Finish : Annealing is done in a controlled atmosphere of cracked ammonia to avoid any oxidation of metal which ensures a bright finish called BA finish. The final surface developed will have a MIRROR type finish. Strips processed through bright annealing line have a brighter luster than material conventionally annealed and pickled.

No8 Finish : This is the most reflective finish obtained by polishing with rotating cloth mops and polishing soaps / paste containing fine abrasives.

Special Finishes

Chequred, Moon Rock, striped Finish

The are typical rolled finishes produced by using an etched roll in the final pass in cold reduction.

Matt Finish : This produced by using a specific rough ground roll during skin passing of 2D Finish material. It offers a matt surface with least reflectivity.

Note : Any specific finish requirements can be supplied with mutual discussion

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Stainless Steel Characteristics & Applications

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Series		Grades	Characteristics	Applications
Austenitic Series	Basic	AISI 304	Widely used, corrosion resistant, heat resistant, good mechanical properties in low temperature, no heat treatment hardening, weak magnetism.	Home Kitchen utensils, industry, medical apparatus, building material and decoration.
		AISI 304L	Better intergranular corrosion resistance, can be heat treated after welding.	Petroleum equipment, chemical equipment, building material, heat resistant parts and the parts are difficult to heat treatment.
	High tensile Strength	AISI 301	Low content of Cr and N than 304 grade, It's strength, hardness and magnetism can be increased by cold working.	Trains, aircrafts, vehicles, springs.
			Improving, intergranular corrosion resistance by decreasing content and increasing N content from 301 grade.	Frame and crust parts of train.
	Deep Drawing	AISI 304Cu	Better formability than 304 grade by addition of cu, suitable for deep drawing.	Domestic appliances.
		AISI 304Ni8.5	Better formability than 304 grade by increasing Ni content, suitable for deep drawing.	Domestic appliances.
		AISI 304Ni9	Better formability than 304 grade by increasing Ni content, suitable for deep drawing.	Domestic appliances.
	High Corrosion Resistance	AISI 316	Better formability than 304 grade in sea water and other corrosive solutions, higher heat-resistance, good cold work hardening and performance, no magnetism.	Chemical industry, food processing equipments, coastal installations.
		AISI 316L	Compare with AISI 316, has better intergnanular corrosion resistance and water heat-resistance, no magnetism.	Chemical industry, food processing equipments used in severe corrosive condition, coastal installations used.
		AISI 321	Intergranular corrosion resistance, heat resistant, heat resistant, good formability and weldability, anti-oxidizing in high temperature.	jet engine parts, heat exchanger, boiler cover, chemical devices.
		AISI 310 / 310S	Excellent high temperature properties with good ductility & weldability.	Used upto 1150°C for continuous service & 1050° for intermittent service in oxidising atmosphere.
Ferretic Series	Basic	AISI 430	Typical ferritic, stainless steel grade, low thermal extensibility, good formability and antioxygenic property.	Heat-resistance utensils, combustion devices, domestic appliance, kitchen utensils category, building decorating material.
	Welded and Corrosion Resistant	AISI 409L	good formability and weldability, antioxygenic property in high temperature.	exhaust system of automobiles
		AISI 436L*	good corrosion resistance, formability and weldability, corrosion resistant to condensat3es in exhaust piping of automobiles.	Automobiles exhaust piping, boiler.
		AISI 444	good corrosion resistance, formability and weldability, compare with 316L, has better stress corrosion resistance and spot corrosion resistance.	Water heater, water tank, automobile, exhaust piping.
		AISI 409 / 409 M	It replaces carbon steels and low alloy steels where higher strength, abrasion resistance, weld-ability and slide-ability are required. It posses good scaling & oxidation resistance at elevated temperature including sulphur bearing atmospheres	Solids handling areas/bunkers, transport(wagons), petrochemical, sugar, agriculture, fishing, mining & quarrying, sewage plants and general engineering industries
Martensite Series	Basic	AISI 403	12% Cr ensuring high degree of cleanliness hard enable by heat treatment.	Reactor components, turbine blades and ring & other highly stressed parts
		AISI 410 / 410S	General purpose corrosion & heat resisting steel; gives maximum corrosion resistance after heat treatment. Hard enable by heat treatment.	Cutlery, Shafts, Cracking tower, babbler caps & trays for oil & gas industry, beater bars for paper & pulp industries etc.
		AISI 420J2*	High post-quenching strength, suitable for wearable parts.	Knife, pipe nozzles, valves, metal rules and cooking utensils.

*Austenitic Stainless Steels are not Hard enable by heat treatment. This can be Hardened by cold work only.



Steel India Co.
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Equivalent International Stainless Steel Grades

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	Grades	UNS Designation	USA - Canada / AISI - ASTM - ASME	INDIA / IS Letter Symbol	European	Chinese	GERMANY / DIN Designation	Japan/ JIS	GOST
Austenitic Cr-Mn	201	S20100	201	X10Cr17Mn6Ni4N20	-	-	X12CrMnNiN17-7-5	SUS201	-
	201L	S20103	201L	-	1.4371	-	X2CrMnNiN17-7-5	-	-
	201LN	S20153	201LN	-	-	-	-	-	-
	202	S20200	202	X10Cr18Mn9Ni5	-	-	X12CrMnNiN18-9-5	SUS202	-
	204 Cu	S20430	-	-					
Austenitic Cr-Ni	301	S30100	301	X10Cr17Ni7	1.4310	1Cr17Ni7	X12CrNi17-7	SUS301	-
	301L	S30103	301L	-	-	-	-	-	-
	301LN	S30153	301LN	-	1.4318	-	X2CrNiN18-7	-	-
	304	S30400	304	X04Cr19Ni9	1.4301	0Cr18Ni9	X5CrNi18-10	SUS304	-
	304H	S30409	304H	-	-	-	-	-	-
	304L	S30403	304L	-	1.4307	-	X2CrNi18-9	SUS304L	-
	304LN	S30453	304LN	-	1.4311	-	X2CrNiN18-10	SUS304LN	-
	309	S30900	309	X15Cr24Ni13	1.4828	-		-	-
	309S	S30908	309S	-	1.4833	1Cr23Ni13	X7CrNi23-14	SUS309S	-
	310	S31000	310	X20Cr25Ni20	-	-	X15CrNiSi25-20	SUH310	20Ch25N20S2
	310S	S31008	310S	-	1.4845	0Cr25Ni20	X12CrNi25-21	SUS310S	20Ch23N18
	316	S31600	316	X04Cr17Ni12Mo2	1.4401	0Cr17Ni12Mo2	X5CrNiMo17-12-2	SUS316	-
	316L	S31603	316L	X02Cr17Ni12Mo2	1.4404	00Cr17Ni14Mo2	X2CrNiMo17-13-2	SUS316L	-
	316LN	S31653	316LN	-	1.4429	-	X2CrNiMoN17-11-2	SUS316LN	-
	316Ti	S31635	316Ti	X04Cr17Ni12Mo2Ti	1.4571	0Cr18Ni12Mo2Ti	X6CrNiMoTi17-12-2	SUS316Ti	10Ch17N13M2T
	317	S31700	317	-	-	-	-	-	-
	317L	S31703	317L	-	1.4438	00Cr19Ni13Mo3	X2CrNiMo18154	SUS317L	-
	317LN	S31753	317LN	-	-	-	-	-	-
	31727	S31727	-	-	-	-	-	-	-
	321	S32100	321	X04Cr18Ni10Ti	1.4541	0Cr18Ni10Ti	X6CrNiTi18-10	SUS321	08Ch18N10T
	347	S34700	347	X04Cr18Ni10Nb	1.4550	0Cr18Ni11Nb	X6CrNiNb18-10	SUS347	08Ch18N12B
Martensitic	410	S41000	410	X12Cr12	1.4006	1Cr12	X12Cr13	SUS410	-
	415	S41500	-	-	1.4313	-	X3CrNiMo13-4	-	-
	420	S42000	420	X20Cr13	1.4021	-	X20Cr13	SUS420JI	-
	431	S43100	431	-	1.4057	1Cr17Ni2	X17CrNi15-2	-	20Ch17N2
Ferritic	405	S40500	405	X04Cr12	1.4002	0Cr13Al	X6CrAl13	SUS405	-
	409	S40900	409	-	1.4512	-	X2CrTi12	SUH409	-
	409RC	-	-	-	-	-	-	-	-
	410S	S41008	410S	-	1.4000	0Cr13	X6Cr-13	SUS403	-
	430	S43000	430	X07Cr17	1.4016	1Cr17	X6Cr-17	SUS430	-
	430Ti	-	-	-	-	-	X3CrTi17	SUS430LX	-
	436	S43600	436	-	-	-	-	-	-
	436L	S43932	436L	-	-	-	-	SUS436L	-
	439	S43035	439	-	-	00Cr18Ti	X3CrTi17	-	-
	441	S43940	-	-	1.4509	-	X2CrTiNb18	-	-
Ferritic + Martensitic									
	409M	-	-	-	-	-	-	-	-
Duplex (Austenitic+ Ferritic)									
	2205	S32205	2205	-	-	-	-	-	-
	31803	S31803	-	-	1.4462	-	X2CrNiMoN 22-5-3	SUS329J3L	-
	2304	S32304	2304	-	1.4362	-	X2CrN N 23-4	-	-

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Mechanical Properties of Stainless Steel

Grade	UTS N/mm² (Min.)	0.2% Proof Stress N/mm² (min.)	% Elongation on 50% GL (Min.)	Hardness RB (max)
301	515	205	40	95
304	515	205	40	92
304L	485	170	40	92
310S	515	205	40	95
316	515	205	40	95
316L	485	170	40	95
321	515	205	40	95
409	380	205	20	88
409M	450	275	20	90
410S	415	205	22	89
430	450	205	22	89

Physical Properties : of Stainless Steel

Property	301	304	316	310S	430	409
Density (gm/cm³)	7.9	7.9	8.0	7.9	7.7	7.7
Modulus of Elasticity (Kg/mm²)	19700	19700	19700	20300	20300	20300
Specific Heat Capacity (Cal/gm/°C)	0.12	0.12	0.12	0.12	0.11	0.11
Thermal Conductivity (Cal/cm²/sec/°C/cm at 100°C)	0.039	0.039	0.0373	0.033	0.0625	0.0595
Specific Electrical Resistance (mW/cm³)	72	72	74	80	60	57
Melting Range (°C)	1400 - 1420	1400-1455	1370-1400	1400-1455	1430-1510	1430-1510
Magnetic	Non Magnetic	Non Magnetic	Non Magnetic	Non Magnetic	Ferro Magnetic	Ferro Magnetic



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Chemical Composition of Stainless Steel

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	Grades	W.Nr.	C Max	Mn Max	P Max	S Max	Si Max	Cr	Ni	Mo	Other Element	Nr. Equ. Specification I.S.	EN
AUSTENITIC CR-MN	SSLNA	-	0.13	10.5	0.75	0.30	0.75	14.0/15.5	0.20/1.0				
	SSLNQ	-	0.12	10.5	0.075	0.015	0.75	14.5/15.5	0.30/1.0				
	SSLN1	-	0.12	10.0	0.075	0.030	0.75	14.0/16.0	0.90/1.70				
	SSLN4	-	0.12	8.0	0.070	0.030	0.75	14.0/17.0	3.80/4.70				
	204 CU	-	0.15	6.5-9.0	0.06	0.03	1	15.5/17.5	1.5-3.5	-	Cu-2-4 N-0.05-0.25	-	-
	301	1.4310	0.15	2.0max	0.045	0.040	1.0	16.0/18.0	6.0/8.0	-	-	10Cr17Ni7	-
	302	1.3410	0.15	2.0	0.045	0.030	1.0	17.0/19.0	8.0/10.0	-	E-4-3-4%	07Cr18Ni9	EN-58A
	302HQ	1.4567	0.03	2.0	0.045	0.03	1	17.0/19.0	9.0/10.0	-	CU:3-4.0	-	-
	303	1.4305	0.15	2.0	0.045	0.15min	1.0	17.0/19.0	8.0/10.0	-	E-4-1%Max	15Cr18Ni9	EN-58M
	303 CU	-	0.15	2.0	0.20	0.15	1	17.0/19.0	8.0/1.0	-	Cu-1.5-3.5	-	-
	304 Cu2	-	0.045	2.0	0.045	0.03	1	18.0/20.0	8.0-10.5	-	Cu-2.3-3.0	-	-
	304	1.4301	0.08	2.0	0.045	0.030	1.0	18.0/20.0	8.0/10.0	-	-	04Cr18Ni10	EN-58E
	304L	1.4307	0.030	2.0	0.045	0.030	1.0	18.0/20.0	8.0/12.0	-	-	02Cr18Ni11	-
	304LN	1.4331	0.03	2.00	0.045	0.03	1	18.0/20.0	8.0-10.5	-	N-0.10-0.15	-	-
	304H		0.05	2.0	0.045	0.03	1	18.0/20.0	8.5-9.5	-	CU:2-2.50	-	-
	308	1.4303	0.08	2.0	0.040	0.030	1.0	18.0/21.0	10.0/12.0	-	-	-	-
	309	1.4829	0.20	2.0max	0.045	0.030	1.0	22.0/24.0	12.0/15.0	-	-	20Cr24Ni12	-
	309LER		0.03	1.5/2.5	0.02	0.015	0.5	23.0/25.0	12.0/14.0	-	-	-	-
	309S	1.4833	0.08	2.0	0.045	0.030	1.0	22.0/24.0	12.0/15.0	-	-	-	-
	310	14.845	0.25	2.0	0.045	0.030	1.50	24.0/26.0	19.0/22.0	-	-	10Cr25Ni12	-
	310S	1.4845	0.08	2.0	0.045	0.030	1.50	24.0/26.0	19.0/22.0	-	-	-	-
	314	1.4841	0.25	2.0	0.040	0.030	1.5	25.0/26.0	19.0/22.0	-	-	-	-
AUSTENITIC CR-NI	316	1.4401	0.08	2.0	0.045	0.030	1.0	16.0/18.0	10.0/14.0	2.0/3.0	-	04Cr17Ni12Mo2	EN-584
	316H	-	0.04-0.10	2.0	0.045	0.030	1	16.0/18.0	10.0/14.0	2.0-3.0	-	-	-
	316L	1.4404	0.030	2.0	0.045	0.030	1.0	16.0/18.0	10.9/14.0	2.0/3.0	-	03Cr17Ni12Mo2	-
	316LN	1.4406	0.03	2.0	0.045	0.03	1	16.0/18.0	10.0/14.0	2.0-/3.0	N-0. 10-0.30	-	-
	316Ti	1.4571	0.080	2.0	0.045	0.030	1.0	16.0/18.0	10.0/14.0	2.0/3.0	Ti5xCmin	-	-
	317	1.4449	0.08	2.0	0.045	0.030	1.0	18.0/20.0	11.0/15.0	3.0/4.0	-	-	-
	317L	1.4438	0.030	2.0	0.045	0.030	1.0	18.0/20.0	11.0/15.0	3.0/4.0	N:0,10/0.22	-	-
	317LN		0.03	2.0	0.045	0.03	1	18.0/20.0	11.0/15.0	3.0/4.0	N:0,10/0.22	-	-
	321	1.4541	0.08	2.0	0.045	0.030	1.0	17.0/19.0	9.0/12.0	-	Ti5xCmin	04Cr18Ni10Ti20	EN-58C
	321H	-	0.04-0.10	2.0	0.045	0.030	1	17.0/19.0	9.0/12.0	-	Ti-4x(C+N)-0.70	-	-
	347	1.4550	0.08	2.0	0.045	0.030	1.0	17.0/19.0	9.0/12.0	-	Nb/Ta10xCmin	04Cr18Ni10Nb-40	EN-58G
FERRITIC / MARTENSITIC	409M	1.4003	0.03	1.0	0.03	0.03	1.0	10.5/12.5	1	-	Ti-0.75max	-	-
	410		0.15	1.00	0.04	0.03	1.0	11.50/13.5	0.60	-	-	12Cr13	En-56A
	416	1.4005	0.15	1.25	0.06	0.15min	1.0	12.0/14.0	1.25/2.50	-	-	-	-
	420	1.4021	0.15min	1.0	0.04	0.03	1.0	12.0/14.0	0.06	-	-	-	En-56C&D
	430	1.4061	0.12	1.0	0.04	0.03	1.0	16.0/18.0	0.06	-	-	07Cr17	En-60
	430L		0.03	1.0	0.04	0.03	1.0	16.0/18.0	0.06	-	-	-	-
	430F		0.12	1.25	0.06	0.15min	1.0	16.0/18.0	0.60	-	-	-	-
	431	1.4057	0.2	1.0	0.04	0.03	1.0	15.0/17.0	1.25/2.5	-	-	-	En57
	446	1.4749	0.02	1.5	0.04	0.03	1	23.0/27.0	0.6	-	N-25	-	-
	410S	1.4001	0.08	1.0	0.04	0.03	1	11.5/13.5	0.60	-	-	-	-
	440A	1.4109	0.6/0.7	1.0	0.04	0.03	1	16.0/18.0	-	-	-	-	-
	440B	1.4112	0.75-0.95	1.0	0.04	0.03	1	16.9/18.0	-	0.75	-	-	-
		440C	1.4125	0.95-1.2	1.0	0.04	0.03	1	16.9/18.0	-	0.75	-	-
DUPLEX	1905	0.03	1.2/1.8	0.04	0.03	1.2/2	18.0/19.0	4.3/5.2	2.5/3	N:0.5/0.10	-	-	
	2205	0.03	2	0.03	0.02	1.0	21.0/23.0	4.5/6.5	2.5/3.5	N:0.8/0.20	-	-	
	2506	0.08	1.0	0.04	0.03	0.75	26.0/28.0	4/5	1.3/2	-	-	-	



Dimensional Tolerances of Stainless Steel

Normal Tolerances on Thickness (Width upto 1500 mm)
Cold Rolled

Thickness (mm)	Tolerance (±) mm
0.30 to < 0.60	0.05
0.60 to < 0.80	0.07
0.80 to < 1.00	0.09
1.00 to < 1.25	0.10
1.25 to < 1.60	0.12
1.60 to < 2.00	0.15
2.00 to < 2.50	0.17
2.50 to < 3.15	0.22
3.15 to < 4.00	0.25
4.00 to < 6.00	0.25

Normal Tolerances on Thickness (width upto 1500 mm)
Hot Rolled

Thickness (mm)	Tolerance (±) mm
2.00 to < 2.25	0.18
2.25 to < 2.50	0.20
2.50 to < 2.75	0.23
2.75 to < 3.00	0.25
3.00 to < 3.25	0.25
3.25 to < 3.50	0.30
3.50 to < 3.75	0.30
3.75 to < 4.00	0.36
4.00 to < 4.99	0.36
4.99 to < 5.00	0.36

- *For thickness above 5.00 mm the applicable tolerance would be 5-8% of the thickness.*
- *Thickness measurement are taken at 20mm from the edge.*
- *The above tolerances are applicable both for sheets and coils.*

Tolerances on Length

Upto 4 mm thickness	+10 mm
Beyond 4 mm thickness	+30 mm

Note : *The above tolerances are applicable for sheets and plates only.*

Selection of welding processes of stainless steel

Welding Method	Recommended Thickness (mm)	Austentic	Weldability Ferritic	Martensitic
Shield Metal Arc Welding (SWAW)	>0.8	Easy to weld	can be welded with care	Difficult to weld Requires Special Care
Gas Tungsten Arc Welding (GTAW)	<3.0	Easy to weld	can be welded with care	can be welded with care
Gas Metal Arc Welding (GMAW)	>3.0	Easy to weld	can be welded with care	can be welded with care
Submerged Arc Welding (SAW)	>6.0	can be welded with care	can be welded with care	can be welded with care
Resistance Spot welding	<3.0	Easy to weld	easy to weld	can be welded with care
Resistance Seam welding	<3.0	Easy to weld	can be welded with care	Difficult to weld Requires special care

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Condensed ASTM Specification for Stainless Steel Tubing and Piping

Specification	Allowable outside Diameter Variations in mm			Allowable Wall Thickness Variations		Exact Length Tolerances in mm		Testing
	Nominal Diameter	Over	Under	% Over	% Under	Over	Under	
ASTMA - 213 Seamless Boiler Superheater and Heat Exchanger Tubes	Under 25.4	.1016	.1016	+20	- 0	3.175	0	Tension Test Flattening Test Flare Test Hardness Test 100% Hydrostatic Test Refer to ASTMA - 450
	25.4 - 381 incl.	.1524	.1524	+20	- 0	3.175	0	
	38.1 - 50.8 excl.	.2032	.2032	+22	- 0	3.176	0	
	50.8 - 63.5 exc.	.254	.254	+22	- 0	4.76	0	
	63.5 - 76.2 Excl.	.3048	.3048	+22	- 0	4.76	0	
	76.2 - 101.6 incl.	.381	.381	+22	- 0	4.76	0	
ASTM A-249 Welder Boiler Superheater and Exchanger and Condenser Tubes	Under 25.4	.1016	.1016	+10	-10	3.175	0	Tension Test Flattering Test Flare Test Reverse Bend Test Hydrostatic Test 100% Hydrostatic Test Reverse Flattening Test Refer to ASTM A-450 wherever applicable
	25.4 - 381 incl.	.1524	.1524	+10	-10	3.175	0	
	38.1 - 50.8 excl.	.2032	.2032	+10	-10	3.175	0	
	50.8 - 63.5 excl.	.254	.254	+10	-10	4.76	0	
	63.5 - 76.2 exc.	.3048	.3048	+10	-10	4.76	0	
	76.2 - 101.6 incl.	.381	.381	+10	-10	4.76	0	
ASTM A-269 Seamless & Welded Tubing for General Service	Upto 12.7	.13	.13	+15	-15	3.2	0	Flare Test Flange Test (Welded only) Hardness Test Reverse Flattening Test (welded only) 100% Hydrostatic Test Refer to ASTM A-269
	12.7 - 38.1 excl.	.13	.13	+10	-10	3.2	0	
	38.1 - 88.9 excl.	.25	.25	+10	-10	4.8	0	
	88.9 - 139.7 excl.	.38	.38	+10	-10	4.8	0	
	139.7 - 203.2 excl.	.76	.76	+10	-10	4.8	0	
ASTM A-312 Seamless and Welded Austenitic Pipe	13.72 - 48.26	+0.40	-0.79	-12.5%				Tension Test Flattening Test 100% Hydrostatic Test
	60.33 - 114.3	+0.79	-0.79					
	141.3 - 219.08	+1.59	-0.79					
	168.28 - 219.08	+1.59	-0.79					
	273.05 - 373.85	+2.38	-0.79					
ASTM A-358 Welded Austenitic pipe	219.08 - 750 mm	± 0.5%		-0.3 mm				Refer to ASTM A-530
ASTM A-409 Welded austenitic pipe	355.6 - 750 mm	±0.2% to ±0.4%		-0.46 mm				Refer to ASTM A-530

Theoretical Mass

Weight (in kg/mtr) & Thickness of SS Pipe & Tube for Boiler, Heat Exchangers.

Size (inch)	O.D. (mm)	10G (3.25)	12G (2.64)	14G (2.03)	16G (1.62)	18G (1.21)	20G (0.91)	22G (0.71)
¼	6.35	0.025	0.232	0.216	0.192	0.152	0.122	0.099
5/16	7.93	0.378	0.349	0.295	0.262	0.199	0.159	0.126
3/8	9.52	0.505	0.448	0.442	0.322	2.659	0.192	0.152
½	12.70	0.751	0.664	0.521	0.445	0.349	0.262	0.209
¾	19.05	1.283	1.033	0.850	0.714	0.535	0.412	0.322
1	25.40	1.798	1.475	1.166	0.977	0.724	0.555	0.435
1 ¼	31.82	2.313	1.927	1.489	1.246	0.914	0.664	-
1 ½	38.10	2.828	2.326	1.801	1.502	1.103	-	-
1 ¾	44.45	3.390	2.765	2.147	1.761	1.296	-	-
2	50.80	3.859	3.191	2.436	2.017	1.485	-	-
2 ¼	57.15	4.371	3.606	2.752	2.283	1.675	-	-
2 ½	63.50	4.892	4.022	3.071	2.632	1.868	-	-
2 ¾	69.85	5.418	4.447	3.311	2.815	2.057	-	-
3	76.20	5.923	4.853	3.706	3.071	2.247	-	-
3 ½	88.90	6.953	5.710	4.341	3.596	2.629	-	-
4	101.60	7.987	6.551	4.976	4.118	3.008	-	-
4 ½	114.30	9.018	7.392	5.344	4.643	3.390	-	-
5	127.00	10.048	8.233	6.245	5.165	3.769	-	-
5 ½	139.70	11.168	9.074	6.880	5.694	4155	-	-
6	152.40	12.145	9.915	7.515	6.232	4.533	-	-
6 ½	165.10	13.212	10.822	8.150	6.741	4.902	-	-

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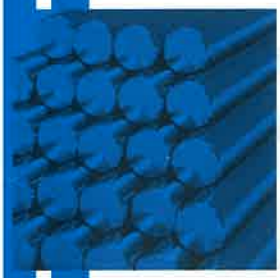
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STAINLESS STEEL PIPE DIMENSION STANDARD SCHEDULE SIZE IN KG. PER MTR.
(As per ANSI B 36. 19-1965)

NOMINAL BORE		Outside Diameter	Schedule 5 S		Schedule 10 S		Schedule 20 S		Schedule 40 S		Schedule 80 S		Schedule 160 S	
MM	INCH	MM	WT MM	WEIGHT (KG/MT)	WT MM	WEIGHT (KG/MT)	WT MM	WEIGHT (KG/MT)	WT MM	WEIGHT (KG/MT)	WT MM	WEIGHT (KG/MT)	WT MM	WEIGHT (KG/MT)
3	1/8	10.3	1.24	0.276	1.24	0.28	1.5	0.33	1.73	0.37	2.41	0.47	-	-
6	1/4	13.7	1.24	0.390	1.65	0.49	2.00	0.58	2.24	0.631	3.02	0.80	-	-
10	3/8	17.1	1.24	0.490	1.65	0.63	2.00	0.74	2.31	0.845	3.20	1.10	-	-
15	½	21.3	1.65	0.80	2.11	1.00	2.30	1.07	2.77	1.27	3.75	1.62	4.75	1.94
20	3/4	26.7	1.65	1.03	2.11	1.28	2.55	1.52	2.87	1.68	3.91	2.20	5.54	2.90
25	1	33.4	1.65	1.30	2.77	2.09	2.55	1.94	3.38	2.50	4.55	3.24	6.35	4.24
32	1.1/4	42.2	1.65	1.65	2.77	2.70	3.00	2.90	3.56	3.39	4.85	4.47	6.35	5.61
40	1.1/2	48.30	1.65	1.91	2.77	3.11	3.00	3.55	3.68	4.05	5.08	5.41	7.14	7.25
50	2	60.3	1.65	2.40	2.77	3.93	3.00	4.24	3.91	5.44	5.54	7.48	8.74	11.1
65	2.1/2	73.0	2.11	3.69	3.05	5.26	4.00	6.81	5.16	8.63	7.01	11.41	9.53	14.9
80	3	88.9	2.11	4.51	3.05	6.45	4.00	8.37	5.49	11.29	7.62	15.27	11.1	21.30
100	4	114.3	2.11	5.84	3.05	8.36	4.50	12.18	6.02	16.07	8.56	22.32	13.49	33.54
125	5	141.3	2.77	9.47	3.40	11.57	5.00	16.80	5.55	21.8	9.53	31.97	15.88	49.11
150	6	168.3	2.77	11.32	3.40	13.84	6.35	25.36	7.11	28.30	10.97	42.70	18.20	67.56
200	8	219.1	2.77	14.79	3.76	19.96	6.35	33.31	8.18	42.61	12.7	64.64	23.01	111.27
250	10	273.1	3.40	22.63	4.19	27.78	6.35	41.77	9.27	60.50				
300	12	323.9	3.96	31.25	4.57	36.00	6.35	49.7	9.53	73.88				
350	14	355.6	3.96	34.36	4.78	41.3	7.92	67.80	11.13	94.59				
400	16	406.4	4.19	41.56	4.78	47.29	7.92	77.83	12.7	123.30				
450	18	457.0	4.19	46.80	4.78	53.42	7.92	87.71	14.27	155.80				
500	20	508.0	4.78	56.25	5.54	68.61	9.53	117.14	15.09	183.42				
600	24	610.0	5.54	82.47	6.35	94.45	9.53	141.12	17.48	255.41				



Formula - Pipe Weight Kg/Mtr OD - Thick x Thick x 0.02466 = Kg Per Mtr.



Theoretical Mass

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Weight of Round, Square & Hexagonal stainless steel Bars
Size in inches, Weight in kg / Metre

Size (inch)	Round	Square	Hexagon	Size (inch)	Round	Square	Hexagon
1/16	0.0149	0.0196	-	2 7/8	33.2400	42.4142	36.7302
1/8	0.0631	0.0801	0.0754	3	36.2316	46.2036	39.9877
3/16	0.1419	0.1811	0.1509	3 ¼	42.5472	54.2144	45.8379
1/4	0.2519	0.3201	0.2715	3 ½	49.3281	62.8568	54.4803
5/16	0.0466	0.5025	0.4377	3 ¾	56.5080	72.1640	62.5244
3/8	0.5657	0.7213	0.6189	4	64.4191	82.1028	71.3330
7/16	0.7711	0.9825	0.8452	4 ¼	72.8288	92.7063	78.8220
1/2	1.0065	1.2827	1.1165	4 ½	81.6374	103.9414	90.0637
9/16	1.2754	1.6244	1.4037	4 ¾	90.9114	115.7749	101.2191
5/8	1.5725	2.0020	1.7357	5	100.7172	128.2731	111.1911
11/16	1.9049	2.4265	2.1127	5 ¼	111.0880	141.4362	122.5891
3/4	2.2669	2.8878	2.5046	5 ½	121.8910	155.1310	134.5389
13/16	2.6605	3.39048	2.9277	5 ¾	133.2591	169.5904	147.0471
7/8	3.0843	3.9289	3.3971	6	144.9264	184.7479	160.1170
15/16	3.5400	4.5106	3.9090	6 ¼	157.3914	200.5701	173.6956
1	4.0220	5.1256	4.4375	6 ½	170.1888	217.0239	187.8824
1 1/8	5.0923	6.4984	5.6275	6 ¾	183.6510	234.0627	202.6709
1 1/4	6.2956	8.0108	6.9405	7	197.4456	251.7265	217.9147
1 3/8	7.6119	9.7060	8.3631	7 ¼	211.7055	269.9752	233.7603
1 1/2	9.041	11.5342	10.0052	7 ½	226.7300	288.8389	250.2141
1 5/8	10.6368	13.5453	11.7436	7 ¾	241.9207	308.4605	267.1166
1 3/4	12.3320	15.7225	13.6151	8	257.6100	328.6771	284.6175
1 7/8	14.1436	18.0327	15.6327	8½	291.4217	371.0913	321.2878
2	16.1214	20.5090	17.7834	9	331.6022	416.0584	360.2218
2 1/8	18.1822	23.2347	20.0769	9½	364.0245	463.4487	401.4228
2 1/4	20.4426	26.0535	22.4702	10	403.2012	513.5447	444.7312
2 3/8	23.2181	29.1847	25.0962	10½	444.6847	566.2134	490.3066
2 1/2	25.1626	32.0766	27.7886	11	487.6308	621.4450	538.1456
2 5/8	27.7554	35.3341	30.6140	11½	533.4189	679.2427	588.2482
2 3/4	30.4810	38.8243	33.5724	12	580.3704	739.4570	640.4616



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Stainless Steel Flanges

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Types and Application

A flange is a forged or cast ring of steel designed to connect section of pipe or join pipe to a pressure vessel, valve, pump or any other integral flanged assembly.

Flanges are joined to each other by bolting and joined to the piping system by welding or threading.

The basic types of flanges are : **Slip On, Blind, Weldneck, Threaded, Socket Weld, Lap Joint and Plate.**

Flanges are designed to the following pressure rating : 150 lb, 300 lb, 400 lb, 600 lb, 900 lb, 1500 lb and 2500 lb or 10 Bar, 15 Bar, 25 Bar, 40 Bar, 64 Bar, 100 Bar and 150 Bar.

The most common facings machined on flanges are : (a) Raised face (b) Flat face (c) Ring type



Slip on Flanges
The Flange is slipped over the pipe and then welded both inside and outside to provide sufficient strength and prevent leakage. This flange is used in preference to weld necks by many users because of its lower cost and the fact that less accuracy is required while cutting pipe to length.



Blind Flanges
This is flange without a bore and it is used to shut off a piping system or vessel opening. It also permits easy access to vessel or piping system for inspection purpose. Blind flanges can be supplied with or without hubs at the manufacturer's option.



Weld neck flanges
This is designed to be joined to a piping system by butt welding. It is relatively expensive because of its long neck, but is preferred for high stresses to the pipe, reducing stress applications. The neck, or hub, transmits stress concentration at the base of the flange. The gradual transition of thickness from the base of the hub to the wall thickness at the butt weld provides important re-inforcement of the flange. The bore of the flange matches the bore of the pipe, reducing turbulence and erosion.



Threaded Flanges
This is similar to a slip on flange in outline but bore is threaded, thus enabling assembly without welding. This obviously limits its application to relatively low pressure piping systems. The flange may be welded around the joint after method of increasing its applications.



Socket weld flanges
This is similar to a slip-on flanges in outline, but the bore is counter-bored to accept pipe. The diameter of the remaining bore is the same as the inside diameter of the pipe. The flange is attached to the pipe by a fillet weld around the hub of the flange. An optional interval weld may be applied in high stress applications. Its biggest use is in high pressure systems such as hydraulic and steam line.



Lap joint flanges
This is again similar to a slip-on flange, but it has radius at the intersection of the bore and the flange face to accommodate a lap stub end. The face on the stub end forms the gasket face of the flange. The type of the flange is used in application where sections of piping systems need to be dismantled quickly and easily for inspection or replacement.



Plate Flange
This is usually used with a pressed collar or stub end and is placed behind the collar or stub end. It is not weld and thus allows for easy alignment. Also permits use of other materials for the flange is not in direct contact with the liquid.



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Dimensions of Various Stainless Steel Flanges

Table D - BS-10 Flange

Flange designation nominal bore of pipe	Approx. Outside diameter of Steel Pipe H	Diameter of flange D	Pitch Circle diameter (PCD) P	No. of Bolts N	Diameter of bolts O	Thickness T
in	in	in	in	in	in	in
½	27/32	3 ¾	2 5/8	4	½	3/16
¾	1 1/16	4	2 7/8	4	½	3/16
1	1 11/32	4 ½	3 ¼	4	½	3/16
1 ¼	1 11/16	4 ¾	3 7/16	4	½	¼
1 ½	1 29/32	5 ¼	3 7/8	4	½	¼
2	2 3/8	6	4 ½	4	5/8	5/16
2 ½	3	6 ½	5	4	5/8	5/16
3	3 ½	7 ¼	5 ¾	4	5/8	3/8
4	4 ½	8 ½	7	4	5/8	3/8
5	5 ½	10	8 ¼	8	5/8	½
6	6 5/8	11	9 ¼	8	5/8	½
8	8 5/8	13 ¼	11 ½	8	5/8	½
10	10 ¾	16	14	8	¾	5/8
12	12 ¾	18	16	12	¾	¾
14	14	20 ¾	18 ½	12	7/8	7/8
16	16	22 ¾	20 ½	12	7/8	7/8
18	18	25 ¼	23	12	7/8	1
20	20	27 ¾	25 ¼	16	7/8	1 1/8
22	22	30	27 ½	16	1	1 1/8
24	24	32 ½	29 ¾	16	1	1 ¼

Table E - BS-10 Flange

Flange designation nominal bore of pipe	Approx. Outside diameter of Steel Pipe H	Diameter of flange D	Pitch Circle diameter (PCD) P	No. of Bolts N	Diameter of bolts O	Thickness T
in	in	in	in	in	in	in
½	27/32	3 ¾	2 5/8	4	½	¼
¾	1 1/16	4	2 7/8	4	½	¼
1	1 11/32	4 ½	3 ¼	4	½	9/32
1 ¼	1 11/16	4 ¾	3 7/16	4	½	5/16
1 ½	1 29/32	5 ¼	3 7/8	4	½	11/32
2	2 3/8	6	4 ½	4	5/8	3/8
2 ½	3	6 ½	5	4	5/8	13/32
3	3 ½	7 ¼	5 ¾	4	5/8	7/16
4	4 ½	8 ½	7	4	5/8	½
5	5 ½	10	8 ¼	8	5/8	9/16
6	6 5/8	11	9 ¼	8	¾	11/16
8	8 5/8	13 ¼	11 ½	8	¾	¾
10	10 ¾	16	14	12	7/8	7/8
12	12 ¾	18	16	12	7/8	1
14	14	20 ¾	18 ½	12	7/8	1 1/8
16	16	22 ¾	20 ½	12	7/8	1 ¼
18	18	25 ¼	23	16	7/8	1 3/8
20	20	27 ¾	25 ¼	16	7/8	1 ½
22	22	30	27 ½	16	1	1 ¾
24	24	32 ½	29 ¾	16	1 1/16	1 7/8

Table F - BS-10 Flange

½	27/32	3 ¾	2 5/8	4	½	3/8
¾	1 1/16	4	2 7/8	4	½	3/8
1	1 11/32	4 ¾	3 7/16	4	5/8	3/8
1 ¼	1 11/16	5 ¼	3 7/8	4	5/8	½
1 ½	1 29/32	5 ½	4 1/8	4	5/8	½
2	2 3/8	6 ½	5	4	5/8	5/8
2 ½	3	7 ¼	5 ¾	8	5/8	5/8
3	3 ½	8	6 ½	8	5/8	5/8
4	4 ½	9	7 ½	8	5/8	¾
5	5 ½	11	9 ¼	8	¾	7/8
6	6 5/8	12	10 ¼	12	¾	7/8
8	8 5/8	14 ½	12 ¾	12	¾	1
10	10 ¾	17	15	12	7/8	11/8
12	12 ¾	19 ¼	17 ¼	16	1	1 ¼
14	14	21 ¾	19 ½	16	1	1 3/8
16	16	24	21 ¾	20	1	1 5/8
18	18	26 ½	24	20	1 1/8	1 ¾
20	20	29	26 ½	24	1 1/8	2
22	22	31	28 ½	24	1 1/8	2 1/8
24	24	33 ½	30 ¾	24	1 ¼	2 ¼

Table H - BS-10 Flange

½	27/32	4 ½	3 ¼	4	5/8	½
¾	1 1/18	4 ½	3 ¼	4	5/8	½
1	1 11/32	4 ¾	3 7/18	4	5/8	8/16
1 ¼	1 11/16	5 ¼	3 7/8	4	5/8	9/16
1 ½	1 29/32	5 ½	4 1/8	4	5/8	¾
2	2 3/8	6 ½	5	4	5/8	11/16
2 ½	3	7 ¼	5 ¾	8	5/8	¾
3	3 ½	8	6 ½	8	5/8	7/8
4	4 ½	9	7 ½	8	5/8	1
5	5 ½	11	9 ¼	8	¾	1 1/8
6	6 5/8	12	10 ¼	12	¾	1 1/8
8	8 5/8	14 ½	12 ¾	12	¾	1 1/4
10	10 ¾	17	15	12	7/8	1 3/8
12	12 ¾	19 ¼	17 ¼	16	7/8	1 5/8
14	14	21 ¾	19 ½	16	1	1 7/8
16	16	24	21 ¾	20	1	2 1/8
18	18	26 ½	24	20	1 1/8	2 3/8
20	20	29	26 ½	24	1 1/8	2 5/8
22	22	31	28 ½	24	1 1/8	2 3/4
24	24	33 ½	30 ¾	24	1 ¼	3

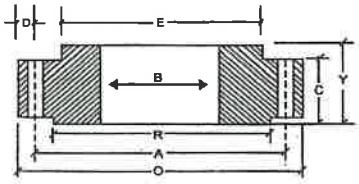
steelindialtd@yahoo.com



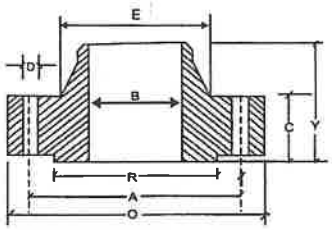
Steel India Co.
(CHENNAI) PVT. LTD.

Dimensions Of Forged Flanges Ansi B 16.5

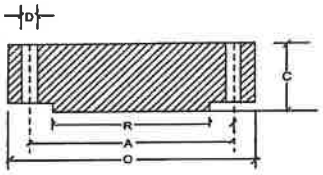
steelindialtd@yahoo.com



SLIP-ON FLANGE



WELDING NECK FLANGE



BLIND FLANGE

ASA 150 CLASS

Nominal Pipe Size (MM)		Flange Dia	Dia of Bolt Circle	Dia of Bolt Holes	No. of Holes	Thk of Flange	Dia of Hub	Length through Hub			Dia of Bore		Dia of R/F	Depth of Socket	Pipe Dia
								S/O & S/W	W/N	L/J	S/O & S/W	L/J			
(MM)	(INCH.)	O	A	D		C	E	Y	Y	Y	B	B	R	F	X
15	1/2	88.9	60.3	15.9	4	11.1	30.2	15.9	47.6	15.9	22.3	22.9	34.9	9.5	21.33
20	3/4	98.4	69.8	15.9	4	12.7	38.1	15.9	52.4	15.9	27.7	28.2	42.9	11.1	26.67
25	1	107.9	79.4	15.9	4	14.3	49.2	17.5	55.6	17.5	34.5	35.0	50.8	12.7	33.40
32	1 1/4	117.5	88.9	15.9	4	15.9	58.7	20.6	57.1	20.6	43.2	43.7	63.5	14.3	42.16
40	1 1/2	127.0	98.4	15.9	4	17.5	65.1	22.2	61.9	22.2	49.5	50.0	73.0	15.9	48.26
50	2	152.4	120.6	19.0	4	19.0	77.8	25.4	63.5	25.4	62.0	62.5	92.1	17.5	60.31
65	2 1/2	177.8	139.7	19.0	4	22.2	90.5	28.6	69.8	28.6	74.7	75.4	104.8	19.0	73.02
80	3	190.5	152.4	19.0	4	23.8	107.9	30.2	69.8	30.2	90.7	91.4	127.0	20.6	88.90
100	4	228.6	190.5	19.0	8	23.8	134.9	33.3	76.2	33.3	116.1	116.8	157.2	23.8	114.30
125	5	254.0	215.9	22.2	8	23.8	163.5	36.5	88.9	36.5	143.8	144.5	185.7	23.8	141.30
150	6	279.4	241.3	22.2	8	25.4	192.1	39.7	88.9	39.7	170.7	171.4	215.9	27.0	168.27
200	8	342.9	298.4	22.2	8	28.6	246.1	44.4	101.6	44.4	221.5	222.2	269.9	31.7	219.07
250	10	406.4	361.9	25.4	12	30.2	304.8	49.2	101.6	49.2	276.3	277.4	323.8	33.3	273.05
300	12	482.6	431.8	25.4	12	31.8	365.1	55.6	114.3	55.6	327.1	328.2	381.0	39.7	323.85
350	14	533.4	476.2	28.6	12	34.9	400.0	57.1	127.0	79.4	359.1	360.2	412.7	41.3	355.60
400	16	596.9	539.7	28.6	16	36.5	457.2	63.5	127.0	87.3	410.5	411.2	469.9	44.4	406.40
450	18	635.0	577.8	31.7	16	39.7	504.8	68.3	139.7	96.8	461.8	462.3	533.4	49.2	457.20
500	20	698.5	635.0	31.7	20	42.9	558.8	73.0	144.5	103.2	513.1	514.3	584.2	54.0	508.00
600	24	812.8	749.3	34.9	20	47.6	663.6	82.5	152.4	111.1	615.9	615.9	692.1	63.5	609.60

All Dimensions are in Millimeters • Flanges except Lap Joint will be furnished with(1.6mm) Raised Face, which is included in Thickness (C) and Lenght through Hub(Y).

ASA 300 CLASS

Nominal Pipe Size (MM)		Flange Dia	Dia of Bolt Circle	Dia of Bolt Holes	No. of Holes	Thk of Flange	Dia of Hub	Length through Hub			Dia of Bore		Dia of R/F	Depth of Socket	Pipe Dia
								S/O & S/W	W/N	L/J	S/O & S/W	L/J			
(MM)	(INCH.)	O	A	D		C	E	Y	Y	Y	B	B	R	F	X
15	1/2	95.2	66.7	15.9	4	14.3	38.1	22.2	52.4	22.2	22.3	22.9	34.9	9.5	21.33
20	3/4	117.5	82.5	19.0	4	15.9	47.6	25.4	57.1	25.4	27.7	28.2	42.9	11.1	26.67
25	1	123.8	88.9	19.0	4	17.5	54.0	27.0	61.9	27.0	34.5	35.0	50.8	12.7	33.40
32	1 1/4	133.3	98.4	19.0	4	19.0	63.5	27.0	65.1	27.0	43.2	43.7	63.5	14.3	42.16
40	1 1/2	155.6	114.3	22.2	4	20.6	69.8	30.2	68.3	30.2	49.5	50.0	73.0	15.9	48.26
50	2	165.1	127.0	19.0	8	22.2	84.1	33.3	69.8	33.3	62.0	62.5	92.1	17.5	60.31
65	2 1/2	190.5	149.2	22.2	8	25.4	100.0	38.1	76.2	38.1	74.7	75.4	104.8	19.0	73.02
80	3	209.5	168.3	22.2	8	28.6	117.5	42.9	79.4	42.9	90.7	91.4	127.0	20.6	88.90
100	4	254.0	200.0	22.2	8	31.8	146.0	47.6	85.7	47.6	116.1	116.8	157.2	23.8	114.30
125	5	279.4	234.9	22.2	8	34.9	177.8	50.8	98.4	50.8	143.8	144.5	185.7	-	141.30
150	6	317.5	269.9	22.2	12	36.5	206.4	52.4	98.4	52.4	170.7	171.4	215.9	-	168.27
200	8	381.0	330.2	25.4	12	41.3	260.3	61.9	111.1	61.9	221.5	222.2	269.9	-	219.07
250	10	444.5	387.3	28.6	16	47.6	320.7	66.7	117.5	95.2	276.3	277.4	323.8	-	273.05
300	12	520.7	450.8	31.7	16	50.8	374.6	73.0	130.2	101.6	327.1	328.2	381.0	-	323.85
350	14	584.2	514.3	31.7	20	54.0	425.4	76.2	142.9	111.1	359.1	360.2	412.7	-	355.60
400	16	647.7	571.5	34.9	20	57.2	482.6	82.5	146.0	120.6	410.5	411.2	469.9	-	406.40
450	18	711.2	628.5	34.9	24	60.3	533.4	88.9	158.7	130.2	461.8	462.3	533.4	-	457.20
500	20	774.7	685.8	34.9	24	63.5	587.4	95.2	161.9	139.7	513.1	514.3	584.2	-	508.00
600	24	914.4	812.8	41.3	24	69.8	701.7	106.4	168.3	152.4	615.9	615.9	692.1	-	609.60

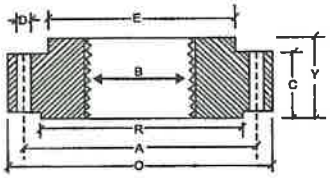
All Dimensions are in Millimeters • Flanges except Lap Joint will be furnished with(1.6mm) Raised Face, which is included in Thickness (C) and Lenght through Hub(Y).



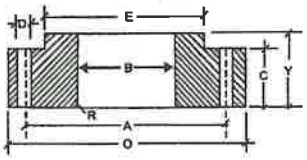
Steel India Co.
(CHENNAI) PVT. LTD.

Dimensions Of Forged Flanges Ansi B 16.5

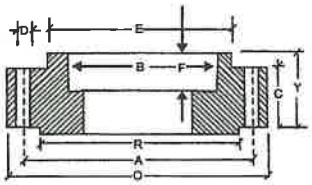
steelindialtd@yahoo.com



THREADED FLANGES



LAP JOINT FLANGES



SOCKET WELD FLANGES

ASA 600 CLASS

Nominal Pipe Size (MM)	Flange Dia	Dia of Bolt Circle	Dia of Bolt Holes	No. of Holes	Thk of Flange	Dia of Hub	Length through Hub			Dia of Bore		Dia of R/F	Depth of Socket	Pipe Dia
							S/O & S/W	W/N	L/J	S/O & S/W	L/J			
	O	A	D		C	E	Y	Y	Y	B	B	R	F	X
15	95.2	66.7	15.9	4	14.3	38.1	22.2	52.4	22.3	22.3	22.8	34.9	9.5	21.33
20	117.5	82.5	19.0	4	15.9	47.6	25.4	57.1	25.4	27.7	28.1	42.9	11.1	26.67
25	123.8	88.9	19.0	4	17.5	54.0	27.0	61.9	26.9	34.5	35.0	50.8	12.7	33.40
32	133.3	98.4	19.0	4	20.6	63.5	28.6	66.7	28.4	43.2	436	63.5	14.2	42.16
40	155.6	114.3	22.2	4	22.2	69.8	31.7	69.8	31.7	49.5	50.0	73.0	15.8	48.26
50	165.1	127.0	19.0	8	25.4	84.1	36.5	73.0	36.5	62.0	62.4	92.1	17.4	60.31
65	190.5	149.2	22.2	8	28.6	100.0	41.3	79.4	41.1	74.7	75.4	104.8	19.0	73.02
80	209.5	168.3	22.2	8	31.8	117.5	46.0	82.5	45.9	90.7	91.4	127.0	-	88.90
100	273.0	215.9	25.4	8	38.1	152.4	54.0	101.6	53.8	116.1	116.8	157.2	-	114.30
125	330.2	2.667	28.6	8	44.4	188.9	60.3	114.3	60.4	143.8	144.5	185.7	-	141.30
150	355.6	292.1	28.6	12	47.6	222.2	66.7	117.5	66.5	170.7	171.4	215.9	-	168.27
200	419.1	349.2	31.7	12	55.6	273.0	76.2	133.3	76.2	221.5	222.2	269.9	-	219.07
250	508.0	431.8	34.9	16	63.5	342.9	85.7	152.4	111.2	276.3	277.4	323.8	-	273.05
300	558.8	488.9	34.9	20	66.7	400.0	92.1	155.6	117.3	327.1	328.2	381.0	-	323.85
350	603.2	527.0	38.1	20	69.9	431.8	93.7	165.1	127.0	359.1	360.1	412.7	-	355.60
400	685.8	603.2	41.3	20	76.2	495.3	106.4	177.8	139.7	410.5	411.2	469.9	-	406.40
450	742.9	654.0	44.4	20	82.6	546.1	117.5	184.1	152.4	461.8	462.3	533.4	-	457.20
500	812.8	723.9	44.4	24	88.9	609.9	127.0	190.5	165.1	513.1	514.3	584.2	-	508.00
600	939.8	838.2	50.8	24	101.6	717.5	139.7	203.2	184.1	615.9	615.9	692.1	-	609.00

All Dimensions are in Millimeters • Flanges except Lap Joint will be furnished with(6.35mm) Raised Face, which is not included in Thickness (C) and Lenght through Hub(Y).

ASA 900 CLASS

Nominal Pipe Size (MM)	Flange Dia	Dia of Bolt Circle	Dia of Bolt Holes	No. of Holes	Thk of Flange	Dia of Hub	Length through Hub			Dia of Bore		Dia of R/F	Depth of Socket	Pipe Dia
							S/O & S/W	W/N	L/J	S/O & S/W	L/J			
	O	A	D		C	E	Y	Y	Y	B	B	R	F	X
15	120.6	82.5	22.2	4	22.2	38.1	31.7	60.3	31.7	22.3	22.8	34.9	9.5	21.33
20	130.2	88.9	22.2	4	25.4	44.4	34.9	69.8	35.0	27.7	28.1	42.9	11.1	26.67
25	149.2	101.6	25.4	4	28.6	52.4	41.3	73.0	41.1	34.5	35.0	50.8	12.7	33.40
32	158.7	111.1	25.4	4	28.6	63.5	41.3	73.0	41.1	43.2	43.6	63.5	14.2	42.16
40	177.8	123.8	28.6	4	31.8	69.8	44.4	82.5	44.4	49.5	50.0	73.0	15.8	48.26
50	215.9	165.1	25.4	8	38.1	104.8	57.1	101.6	57.1	62.0	62.4	92.1	17.4	60.31
65	244.5	190.5	28.6	8	41.3	123.8	63.5	104.8	63.5	74.7	75.4	104.8	19.0	73.02
80	241.3	190.5	25.4	8	38.1	127.0	53.9	101.6	53.8	90.7	91.4	127.0	-	88.90
100	292.1	234.9	31.7	8	44.4	158.7	69.8	114.3	69.8	116.0	116.8	157.2	-	114.30
125	349.2	279.4	35.0	8	50.8	190.5	79.3	127.0	79.2	143.7	144.5	185.7	-	141.30
150	381.0	317.5	31.7	12	55.6	234.9	85.8	139.7	85.8	170.6	171.4	215.9	-	168.27
200	469.9	393.7	38.1	12	63.5	298.4	101.6	162.0	114.3	221.4	222.2	269.9	-	219.07
250	546.1	469.9	38.1	16	69.8	368.3	107.9	184.1	127.0	276.3	277.3	323.8	-	273.05
300	609.6	533.4	38.1	20	79.3	419.1	117.4	200.0	142.7	327.1	328.1	381.0	-	323.85

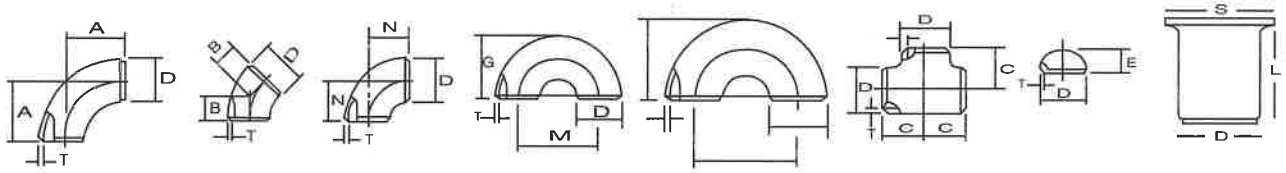
All Dimensions are in Millimeters • Flanges except Lap Joint will be furnished with(6.35mm) Raised Face, which is not included in Thickness (C) and Lenght through Hub(Y).



Steel India Co.
(CHENNAI) PVT. LTD.

Dimensions Of Butt-welding Fitting Ansi B-16.9 / B-16.28

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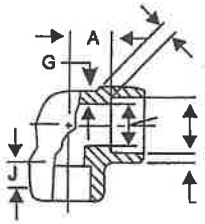


90° Elbow Long Radius | 45° Elbow | 90° Elbow Short Radius | 180° Return Short Radius | 180° Return Long Radius | Tee Equal Tee | Caps | Stub-End

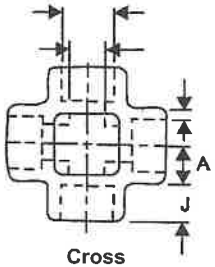
Nominal Pipe Size		Outside Diameter	Center to Face				Back to Face			Center to Center			Length 'L' MSS SP 43 B16.9	
INCH	MM	D	A R=1.5D	B	C	N R=1D	E	F	G	R	M	S	Short L	Long L
1/2	15	21.3	38.00	16.0	25.0	-	25.0	48.0	-	76.0		35.0	50.8	76.2
3/4	20	26.7	29.00	11.0	29.0	-	25.0	43.0	-	57.0		43.0	50.8	76.2
1	25	33.4	38.00	22.0	38.0	25.0	38.0	56.0	41.0	76.0	51.0	51.0	50.8	101.6
1.1/4	32	42.2	48.00	25.0	48.0	32.0	38.0	70.0	52.0	95.0	64.0	64.0	50.8	101.6
1.1/2	40	48.3	57.15	29.0	57.0	38.0	38.0	83.0	62.0	114.0	76.0	73.0	50.8	101.6
2	50	60.3	76.00	35.0	64.0	51.0	38.0	106.0	81.0	152.0	102.0	93.0	63.5	152.4
2.1/2	65	73.0	95.25	44.0	76.0	64.0	38.0	132.0	100.0	191.0	127.0	105.0	63.5	152.4
3	80	88.9	114.30	51.0	86.0	76.0	51.0	159.0	121.0	229.0	152.0	127.0	63.5	152.4
3.1/2	90	101.6	133.35	57.0	95.0	89.0	64.0	184.0	140.0	267.0	178.0	140.0	76.2	152.4
4	100	114.3	152.0	64.0	105.0	102.0	64.0	210.0	159.0	305.0	203.0	157.0	76.2	152.4
5	125	141.3	190.0	79.0	123.0	127.0	76.0	262.0	197.0	381.0	254.0	186.0	76.2	203.2
6	150	168.3	229.0	95.0	143.0	152.0	102.0	313.0	237.0	457.0	305.0	218.0	88.9	203.2
8	200	219.1	305.0	127.0	178.0	203.0	89.0	414.0	313.0	610.0	406.0	270.0	101.6	203.2
10	250	273.1	381.0	159.0	216.0	254.0	102.0	515.0	391.0	762.0	508.0	324.0	127.0	254.0
12	300	323.8	457.0	190.0	254.0	305.0	127.0	619.0	467.0	914.0	610.0	381.0	152.4	254.0
14	350	355.6	533.0	222.0	279.0	356.0	152.0	711.0	533.0	1067.0	711.0	413.0	152.4	305.0
16	400	406.4	610.0	254.0	305.0	406.0	165.0	813.0	610.0	1219.0	813.0	470.0	152.4	305.0
18	450	457.2	686.0	286.0	343.0	457.0	178.0	914.0	686.0	1372.0	914.0	533.0	152.4	305.0
20	500	508.0	762.0	318.0	381.0	508.0	203.0	1016.0	762.0	1524.0	1016.0	584.0	152.4	305.0
22	550	559.0	838.0	343.0	419.0	559.0	229.0	1118.0	838.0	1676.0	1118.0	614.4	152.4	305.0
24	600	610.0	914.0	381.0	432.0	610.0	254.0	1219.0	914.0	1829.0	1219.0	692.0	152.4	305.0
26	650	660.0	991.0	405.0	495.0	660.0	267.0							
28	700	711.0	1067.0	438.0	521.0	771.0	267.0							
30	750	762.0	1143.0	470.0	559.0	762.0	267.0							
32	800	813.0	1219.0	502.0	597.0	813.0	267.0							
34	850	864.0	1295.0	533.0	635.0	864.0	267.0							
36	900	914.4	1372.0	565.0	673.0	914.0	267.0							



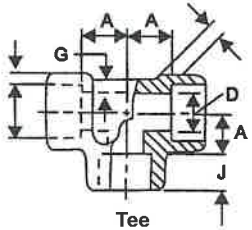
All Dimensions in Millimeters



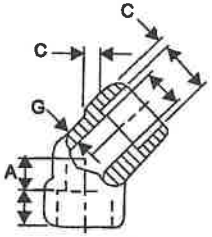
90° deg Elbow



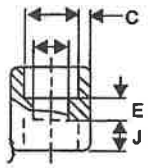
Cross



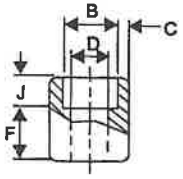
Tee



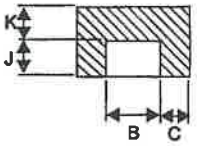
45° deg Elbow



Coupling



Half Coupling



Cap

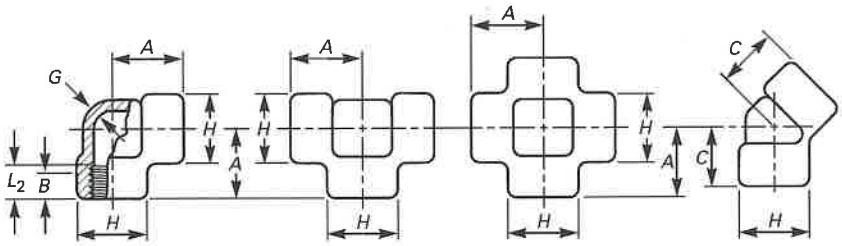
Avg. Min.

Nominal Pipe Size		Socket Bore Diameter B Note 1	Bore Diameter of Fitting D [Note(1)]			Socket Wall Thickness, C [Note (2)]						Body Wall G			Min. Depth of Socket J	Center to Bottom of Socket, A						Laying Lengths		Tolerances			End Wall Thickness K.Min.		
			Class Designation			Class Designation						Class Designation				90° Elbows, Tees& Crosses			45° Elbows,							Class Designation		
						3000		6000		9000		3000	6000	9000		Class Designation						Couplings E	Half Couplings F				3000	6000	9000
NB	INCH		3000	6000	9000	Avg.	Min.	Avg.	Min.	Avg.	Min.	Min.	Min.	Min.		3000	6000	9000	3000	6000	9000			A	E	F			
6	1/8	11.2	7.6	4.8	...	3.18	3.18	3.96	3.43	...	...	2.41	3.15	...	9.5	11.0	11.0	...	8.0	8.0	...	6.5	16.0	1.0	1.5	1.0	4.8	6.4	..
		10.8	6.1	3.2	...	...																							
8	1/4	14.6	10.0	7.1	...	3.78	3.30	4.60	4.01	...	...	3.02	3.68	...	9.5	11.0	13.5	...	8.0	8.0	...	6.5	16.0	1.0	1.5	1.0	4.8	6.4	...
		14.2	8.5	5.6	...	...																							
10	3/8	18.0	13.3	9.9	...	4.01	3.50	5.03	4.37	...	...	3.20	4.01	...	9.5	13.5	15.5	...	8.0	11.0	...	6.5	17.5	1.5	3.0	1.5	4.8	6.4	...
		17.6	11.8	8.4	...																								
15	1/2	22.2	16.6	12.5	7.2	4.67	4.09	5.97	5.18	9.35	8.18	3.73	4.78	7.47	9.5	15.5	19.5	25.5	11.0	12.5	15.5	9.5	22.5	1.5	3.0	1.5	6.4	7.9	11.2
		21.8	15.0	11.0	5.6																								
20	3/4	27.6	21.7	16.3	11.8	4.90	4.27	6.96	6.04	9.78	8.56	3.91	5.56	7.82	12.5	19.0	22.5	28.5	13.0	14.0	19.0	9.5	24.5	1.5	3.0	1.5	6.4	7.9	12.7
		27.2	20.2	14.8	10.3																								
25	1	34.3	27.4	21.5	16.0	5.69	4.98	7.92	6.93	11.38	9.96	4.55	6.35	9.09	12.5	22.5	27.0	32.0	14.0	17.5	20.5	12.5	28.5	2.0	4.0	2.0	9.6	11.2	14.2
		33.9	25.9	19.9	14.4																								
32	1 1/4	43.1	35.8	30.2	23.5	6.07	5.28	7.92	6.93	12.14	10.62	4.85	6.35	9.70	12.5	27.0	32.0	35.0	17.5	20.5	22.5	12.5	30.0	2.0	4.0	2.0.	9.6	11.2	14.2
		42.7	34.3	28.7	22.0																								
40	1 1/2	49.2	41.6	34.7	28.7	6.35	5.54	8.92	7.80	12.70	11.12	5.08	7.14	10.15	12.5	32.0	38.0	38.0	20.5	25.5	25.5	12.5	32.0	2.0	4.0	2.0	11.2	12.7	15.7
		48.8	40.1	33.2	27.2																								
50	2	61.7	53.3	43.6	38.9	6.93	6.04	10.92	9.50	13.84	12.12	5.54	8.74	11.07	16.0	38.0	41.0	54.0	25.5	28.5	28.5	19.0	41.0	2.0	4.0	2.0	12.7	15.7	19.0
		61.2	51.7	42.1	37.4																								
65	2 ½	74.4	64.2	...	...	8.76	7.67	...	...	...	...	7.01	...	...	16.0	41.0	—	...	28.5	...	...	19.0	43.0	2.5	5.0	2.5	15.7	19.0	...
		73.9	61.2	...	...																								
80	3	90.3	79.4	...	...	9.52	8.30	...	...	...	...	7.62	...	...	16.0	57.0	—	...	32.0	...	...	19.0	44.5	2.5	5.0	2.5	19.0	22.4	...
		89.8	76.4	...	...																								
100	4	115.7	108.8	...	...	10.69	9.35	...	...	...	...	8.56	...	...	19.0	66.5	—	...	41.0	...	...	19.0	48.0	2.5	5.0	2.5	22.4	28.4	...
		115.2	100.7	...	...																								

(1) All Dimensions Are In Millimeters
(2) Upper and lower values for each size are the respective maximum and minimum dimensions.
(3) Average of socket wall thickness around periphery shall be no less than listed values. The minimum values are permitted in localized areas.

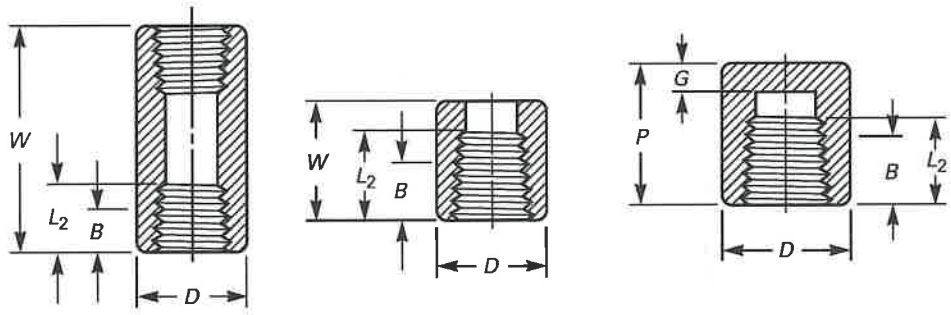
Dimensions Of Threaded Fittings Ansi B 16.11

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90-deg Elbow | Tee | Cross | 45-deg Elbow

Nominal Pipe Size		Center to End Elbows, Tees, Crosses A			Center to End 45 Elbows C			Outside Diameter of Bend H			Minimum wall Thickness G			Min. Length of Thread Note - 1	
NB	INCH	2000	3000	6000	2000	3000	6000	2000	3000	6000	2000	3000	6000	B	L2
6	1/8	21	21	25	17	17	19	22	22	25	3.18	3.18	6.35	6.4	6.7
8	1/4	21	25	28	17	19	22	22	25	33	3.18	3.30	6.60	8.1	10.2
10	3/8	25	28	33	19	22	25	25	33	38	3.18	3.51	6.98	9.1	10.4
15	1/2	28	33	38	22	25	28	33	38	46	3.18	4.09	8.15	10.9	13.6
20	3/4	33	38	44	25	28	33	38	46	56	3.18	4.32	8.53	12.7	13.9
25	1	38	44	51	28	33	35	46	56	62	3.68	4.98	9.93	14.7	17.3
32	1 1/4	44	51	60	33	35	43	56	62	75	3.89	5.28	10.59	17.0	18.0
40	1 1/2	51	60	64	35	43	44	62	75	84	4.01	5.56	11.07	17.8	18.4
50	2	60	64	83	43	44	52	75	84	102	4.27	7.14	12.09	19.0	19.2
65	2 1/2	76	83	95	52	52	64	92	105	121	5.61	7.65	15.29	23.6	28.9
80	3	86	95	106	64	64	79	109	121	146	5.99	8.84	16.64	25.9	30.5
100	4	106	114	114	79	79	79	146	152	152	6.55	11.18	18.67	27.7	33.0



Coupling | Half-Coupling | Cap

Nominal Pipe Size		End to End Couplings W	End to End Caps P		Outside Diameter D		End Wall Thickness min. G		Min. Length of Thread Note - 1	
NB	INCH	3000 and 6000	3000	6000	3000	6000	3000	6000	B	L2
6	1/8	32	19	-	16	22	4.8	-	6.4	6.7
8	1/4	35	25	27	19	25	4.8	6.4	8.1	10.2
10	3/8	38	25	27	22	32	4.8	6.4	9.1	10.4
15	1/2	48	32	33	28	38	6.4	7.9	10.9	13.6
20	3/4	51	37	38	35	44	6.4	7.9	12.7	13.9
25	1	60	41	43	44	57	9.7	11.2	14.7	17.3
32	1 1/4	67	44	46	57	64	9.7	11.2	17.0	18.0
40	1 1/2	79	44	48	64	76	11.2	12.7	17.8	18.4
50	2	86	48	51	76	92	12.7	15.7	19.0	19.2
65	2 1/2	92	60	64	92	108	15.7	19.0	23.6	28.9
80	3	108	65	68	108	127	19.0	22.4	25.9	30.5
100	4	121	68	75	140	159	22.4	28.4	27.7	33.0



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Carbon Steels: Chemical Composition

Grades	% Chemical Composition													Deoxidation
	C	Mn	Si	S	P	Al	Cr	Ni	Cu	Nb	V	Ti	N	
IS 1079 Gr O	0.15 max	0.6 max	-	0.055 max	0.055 max	-	-	-	-	-	-	-	-	Semi Killed / Killed
IS 1079 Gr D	0.12 max	0.5 max	-	0.04 max	0.04 max	-	-	-	-	-	-	-	-	Semi Killed / Killed
IS 1079 Gr DD	0.1 max	0.4 max	-	0.035 max	0.035 max	0.02 min	-	-	-	-	-	-	-	Al Killed
IS 1079 Gr EDD	0.08 max	0.4 max	-	0.03 max	0.03 max	0.02 min	-	-	-	-	-	-	-	Al Killed
IS 2062 E250 A	0.23 max	1.5 max	0.40 max	0.045 max	0.045 max	-	-	-	-	-	-	-	-	Semi Killed / Killed
IS 2062 E250 B	0.22 max	1.5 max	0.40 max	0.045 max	0.045 max	-	-	-	-	-	-	-	-	Killed
IS 2062 E250 C	0.2 max	1.5 max	0.40 max	0.04 max	0.04 max	-	-	-	-	-	-	-	-	Killed
IS 2062 E250Cu C	0.2 max	1.5 max	0.40 max	0.04 max	0.04 max	-	-	-	0.2-0.35	-	-	-	-	Killed
IS 2062 E410	0.20 max	1.6 max	0.45 max	0.045 max	0.045 max	-	-	-	-	-	-	-	-	Killed
IS 2062 E450 D	0.22 max	1.6 max	0.45 max	0.045 max	0.045 max	-	-	-	-	-	-	-	-	Killed
IS 2062 E450 E	0.22 max	1.80 max	0.45 max	0.045 max	0.045 max	-	-	-	-	-	-	-	-	Killed
IS 5986 Fe410	0.20 max	1.20 max	-	0.040 max	0.040 max	-	-	-	-	-	-	-	-	Killed
IS 10748 Gr 1	0.10 max	0.50 max	-	0.040 max	0.040 max	-	-	-	-	-	-	-	-	killed
IRS M 41-97/ SAILCOR	0.1 max	0.25-0.45	0.28-0.72	0.03 max	0.075-0.14	0.08 max	0.35-0.60	0.20-0.47	0.3-0.6	-	-	-	-	Killed
SS 4012A-E34	0.1 max	0.7 max	0.2 max	0.03 max	0.03 max	0.02-0.05	-	-	-	0.055 max	0.095 max	0.045 max	-	Al Killed
SS 4012A-E38	0.1 max	1.0 max	0.4 max	0.03 max	0.03 max	0.02-0.05	-	-	-	0.055 max	0.095 max	0.045 max	-	Al Killed
SAILMA 350/350 Hi	0.25	1.50	0.40	0.055	0.055	0.01 min	-	-	-	-	-	-	-	Killed
SAILMA 410/410Hi	0.25	1.50	0.40	0.055	0.055	0.01 min	-	-	-	-	-	-	-	Killed
SAILMA 450/450 Hi	0.25	1.50	0.40	0.055	0.055	0.01 min	-	-	-	-	-	-	-	Killed

Micro Alloying elements like Nb, V, Ti may be present simply or in combination, the limits are given below:

Grade	Nb, V, Ti (max)
IS 2062 E250 A, B, C, Cu C IS 2062 E410 IS 2062 E450 D, IS 2062 E 450 E	0.25%
SAILMA350 SAILMA 350 Hi	0.20 %
SAILMA410 SAILMA 410 Hi	0.20 %
SAILMA450 SAILMA 450 Hi	0.20 %

Incidental element limit in grade IRSM-41/97/SAILCOR is as follows

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Element	Limit
Mo	0.05 max
V	0.05 max
Al	0.08 max
Nb	0.04 max
Mo + V + Al + Nb	0.15 max

Maximum carbon equivalent for grades is as follows:

Grade	Carbon Equivalent (max)
IS 2062 E250 A	0.42
IS 2062 E250 B	0.41
IS 2062 E250 C	0.39
IS 2062 E410	0.44
IS 2062 E450 D	0.46
IS 2062 E450 E	0.48
IS 5986 Fe410	0.42

MECHANICAL PROPERTIES OF STANDARD GRADES

Grades	YS N/mm^2	UTS N/mm^2	% El (Min) GL= $5.65\sqrt{So}$	Bend Test (t)	Hard R_B	Charpy V-Notch Impact Energy (min)
IS 1079 Gr O	-	-	-	2 t	-	-
IS 1079 Gr D	-	240- 400	25	1 t	-	-
IS 1079 Gr DD	-	260-390	28	Close	-	-
IS 1079 Gr EDD	-	260-380	32	Close	-	-
IS 2062 E250 A	250 min	410 min	23	3 t	-	-
IS 2062 E250 B	250 min	410 min	23	2 t	-	27 J at Room temp See Note
IS 2062 E250 C	250 min	410 min	23	2 t	-	27 J at Room temp See Note
IS2062E250Cu C	250 min	410 min	23	2 t	-	27 J at Room temp See Note
IS 2062 E410	410 min	540 min	23	2 t	-	50 J at Room temp See Note
IS 2062 E450 D	450 min	570 min	20	2 t	-	45 J at Room temp See Note
IS 2062 E450 E	450 min	590 min	20	2 t	-	45 J at Room temp See Note
IS 5986 Fe410	255 min	410-520	24 for t >3/0 mm *	2t	-	-
IS 10748 Grade I	170 min	290 min	30	T	-	-
SS 4012A E-34	334-412	392-490	27	Close (For 't'<7.0 mm) 0.5 t(For 't'=7.0)	-	-
SS4012A E-38	373-461	442-559	25	0.5 t	-	-
IRS M 41-97/ SAILCOR	340 min	480 min	22	1 t	-	-

't' : Nominal thickness of test piece, * : Elongation 15 min N in 80 mm GL for $t \leqslant 3.0$ mm
Note: For grade IS 2062 E250 B,IS 2062 E250 C,E410,E450 D,E450 E Impact Test shall be certified for product thickness of 12 mm or more. The testing temperature will be room temperature unless otherwise specified in the order.

WEIGHT OF RODS IN KILOGRAMS PER LINER FOOR

Size			Brass			Aluminium			Stainless Steet / Alloy Steel		
Inch	Fraction of inch	mm	Round	Hex	Sq.	Round	Hex	Sq.	Round	Hex	Sq.
1/8"	0.125	3.173	0.020	0.023	0.026	—	—	—	—	—	—
3/16"	0.187	4.762	0.046	0.051	0.059	0.015	0.017	0.020	0.043	0.047	0.054
1/4"	0.250	6.350	0.082	0.91	0.104	0.030	0.035	0.037	0.076	0.084	0.097
5/16"	0.312	7.937	0.129	0.142	0.164	0.042	0.047	0.054	0.119	0.132	0.151
3/8"	0.375	9.52	0.185	0.224	0.235	0.061	0.068	0.078	0.171	0.188	0.218
7/16"	0.437	11.112	0.252	0.278	0.321	0.083	0.092	0.106	0.233	0.257	0.296
1/2"	0.500	12.700	0.329	0.363	0.419	0.109	0.120	0.138	0.305	0.336	0.386
9/16"	0.562	14.287	0.416	0.461	0.529	0.138	0.152	0.175	0.384	0.424	0.490
5/8"	0.625	15.875	0.516	0.566	0.652	0.170	0.188	0.217	0.473	0.520	0.604
11/16"	0.687	17.462	0.621	0.684	0.789	0.206	0.227	0.262	0.587	0.647	0.760
3/4"	0.750	19.050	0.739	0.817	0.940	0.245	0.270	0.312	0.687	0.736	0.870
13/16"	0.812	20.637	0.867	0.958	1.104	0.287	0.317	0.366	0.800	0.882	1.021
7/8"	0.875	22.225	1.008	1.109	1.282	0.333	0.368	0.424	0.931	1.038	1.184
15/16"	0.937	23.812	1.154	1.273	1.471	0.383	0.422	0.486	1.067	1.176	1.359
1"	1.00	25.400	1.314	1.451	1.675	0.436	0.481	0.555	1.223	1.350	1.546
1.1/8"	1.125	28.575	1.665	1.834	2.122	0.553	0.608	0.705	1.569	1.730	1.950
1.1/4"	1.250	31.750	2.053	2.268	2.619	0.680	0.748	0.867	1.888	2.080	2.415
1.3/8"	1.375	34.925	2.487	2.742	3.167	0.980	1.082	1.248	2.760	3.042	3.478
1.1/2"	1.500	38.100	2.961	3.262	3.167	0.821	0.907	1.082	2.318	2.554	2.922
1.5/8"	1.625	41.275	3.742	3.833	4.426	1.152	1.272	1.466	3.205	3.540	4.072
1.34"	1.750	44.450	4.029	4.444	5.129	1.352	1.474	1.720	3.738	4.120	4.756
1.7/8"	1.875	47.625	4.627	5.101	5.890	1.529	1.692	1.942	4.300	4.740	5.475
2"	2.000	50.800	5.266	5.804	6.703	1.742	1.919	2.220	4.905	5.415	6.224
2.1/4"	2.250	57.150	6.662	7.346	8.482	2.204	2.433	2.802	6.210	6.850	7.910
2.1/2"	2.500	63.500	8.225	9.067	10.472	2.722	3.005	3.462	7.614	8.400	9.700
2.3/4	2.750	69.850	10.199	11.225	12.955	3.003	3.315	3.820	9.282	10.20	11.820
3	3.000	76.200	12.454	13.267	15.341	3.919	4.327	4.990	10.996	12.210	14.000
3.1/2	3.500	88.900	16.364	18.040	20.824	5.408	5.970	6.884	14.946	16.500	19.020
4	4.000	101.600	21.477	23.210	27.358	6.968	7.693	8.870	19.619	21.620	25.000

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Steel India Co.
(CHENNAI) PVT. LTD.

BEAMS / CHANNELS / ANGLES / PLATE & ROUND

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M.S BEAMS			M.S CHANNELS		
Size In mm	Weight in Kgs per Feet	Weight in Kgs per Mtr.	Size In mm	Weight in Kgs per Feet	Weight in Kgs per Mtr.
100x50	2.432	7.979	75x40	2.172	7.126
125x27	4.023	13.199	100x50	2.925	9.597
150x75	4.571	14.997	125x65	3.992	13.098
175x85	5.943	19.499	150x75	5.120	16.799
200x125	7.893	25.897	175x75	5.973	19.597
250x125	11.612	38.099	200x75	6.796	22.298
300x140	14.782	48.50	250x80	9.326	30.599
350x140	16.125	52.906	300x90	11.063	36.298
400x140	19.080	62.601	400x100	15.270	50.300
450x150	22.280	73.100	M.S.ANGLES		
500x180	26.730	87.700	20x20x3	0.274	0.899
600x210	37.370	122.600	25x25x3	0.335	1.099
			25x25x3	0.578	1.798

M.S ANGLES			M.S ANGLES		
Size In mm	Weight in Kgs per Feet	Weight in Kgs per Mtr.	Size In mm	Weight in Kgs per Feet	Weight in Kgs per Mtr.
31x31x3	0.390	1.280	75x75x8	2.712	8.898
35x35x5	0.792	2.599	75x75x10	3.352	10.998
37x37x3	0.518	1.700	80x80x6	2.224	7.297
40x40x3	0.548	1.798	80x80x8	2.925	9.697
40x40x5	0.915	3.002	90x90x6	2.499	8.199
40x40x6	1.066	3.498	90x90x8	3.292	10.801
50x50x5	1.158	3.799	90x90x10	4.084	13.400
50x50x6	1.372	4.502	100x100x6	2.804	9.200
60x60x6	1.645	5.397	100x100x8	3.687	12.097
65x65x6	1.767	5.798	100x100x10	4.545	14.912
65x65x8	2.346	7.697	100x100x12	5.345	17.701
65x65x10	2.864	9.397	110x110x12	5.395	17.700
75x75x6	2.072	6.798	130x130x10	6.004	19.699

M.S AMGLES			M.S ROUND		
Size In mm	Weight in Kgs per Feet	Weight in Kgs per Mtr.	Size In mm	Weight in Kgs per Feet	Weight in Kgs per Mtr.
150X150X10	6.950	22.803	6	0.067	0.200
150X150X12	8.290	27.803	8	0.119	0.390
150X150X12	10.911	35.799	10	0.189	0.620
150X150X20	13.441	44.100	12	0.271	0.890
M.S WINDOW SECTION			16	0.482	1.581
"z" (F-7)	0.45	1.476	20	0.753	2.471
4B	0.750	2.461	25	1.173	3.849
3/4" T Angle	0.275	0.902	32	1.923	6.309
1"T Angle	0.350	1.148	36	2.435	7.989
11/2" T Angle	1.100	3.610	40	3.005	9.859
2"T Angle	1.450	4.760	45	3.745	12.287
2.1/2" T Angale	2.100	6.900	50	4.696	15.403
3"T Angle	2.800	9.200	53	5.275	17.307

M.S ROUND			M.S PLATE		
Size In mm	Weight in Kgs per Feet	Weight in Kgs per Mtr.	Size In mm	Weight in Kgs per Feet	Weight in Kgs per Mtr.
56	5.895	19.341	4	3.00	32.900
60	6.760	22.180	5	3.800	40.000
63	7.458	24.470	6	4.500	48.000
70	10.563	34.657	8	6.000	54.000
80	12.026	39.457	10	7.500	80.000
90	15.220	49.937	12	9.000	96.000
100	18.793	61.660	16	12.000	128.000
110	22.736	74.597	20	15.000	160.00
125	29.362	96.3325	25	18.750	200.00
M.S. GATE CHANNEL			M.S.CHEQURED PLATE		
Coollapseble		1	3,100	34.000	
Gate	0.180	0.590	5	3.900	42.000
41x32	1.456	4.790	6	4.750	50.000

IS - 4923 (GR. YST 240) SPECIFICATION OF STRUCTURE (SQUARE & RECTANGLE)

steelindialtd@yahoo.com

(SHS) Square		
D x B Size	t mm	W kg/m
25 x 25	1.6	1.12
	2.0	1.36
	2.6	1.69
	3.2	1.98
32 x 32	2.0	1.80
	2.6	2.26
	3.2	2.69
38 x 38	2.0	2.18
	2.6	2.75
	3.2	3.29
	4.0	3.95
40 x 40	2.0	2.92
	2.6	3.21
	3.2	3.49
	4.0	4.20
49.5 x 49.5	2.6	3.69
	2.9	4.07
	3.6	4.93
	4.5	5.95
60 x 60	2.6	4.55
	2.9	5.03
	3.2	5.50
	4.0	6.71
	4.8	7.85
72 x 72	3.2	6.71
	4.0	8.22
	4.8	9.66
80 x 80	3.2	7.51
	4.0	9.22
	4.8	10.87
91.5 x 91.5	3.8	9.67
	4.5	11.88
	5.4	14.01
100 x 100	4.0	11.73
	5.0	14.41
	6.0	16.98
113.5 x 113.5	4.8	15.92
	5.4	17.74
132 x 132	4.8	16.71
	5.4	20.88
150 x 150	4.0	18.01
	5.0	22.26
	6.0	26.40
	7.0	30.44
	8.0	44.38
180 x 180	4.0	21.78
	5.0	26.97
	6.0	32.05
	7.0	33.03
	8.0	41.90
220 x 220	4.0	26.61
	5.0	33.25
	6.0	39.59
	7.0	45.83
	8.0	51.96
250 x 250	4.0	30.57
	5.0	37.96
	6.0	45.24
	7.0	52.42
	8.0	59.50

(RHS) Rectangle		
D x B Size	t mm	W kg/m
50 x 25	2.0	2.15
	2.6	2.71
	3.2	3.24
	4.0	3.88
60 x 40	2.6	3.73
	2.9	4.12
	3.6	4.98
	4.5	6.02
66 x 33	2.6	3.69
	2.9	4.07
	3.6	4.93
	4.5	5.95
80 x 40	2.6	4.55
	2.9	5.03
	3.2	5.50
	4.0	6.71
	4.8	7.85
96 x 48	3.2	6.71
	4.0	8.22
	4.8	9.66
100 x 50	3.2	7.01
	4.0	8.59
122 x 61	3.6	9.67
	4.5	11.88
	5.4	14.01
120 x 60	3.2	8.51
	3.6	9.50
	4.5	11.67
145 x 82	4.8	15.92
	5.4	17.74
172 x 92	4.8	18.71
	5.4	20.88
200 x 100	4.0	18.01
	5.0	22.26
	6.0	24.40
	7.0	30.44
	8.0	34.38
220 x 140	4.0	21.78
	5.0	26.97
	6.0	32.05
	7.0	37.03
240 x 120	4.0	21.78
	5.0	26.97
	6.0	32.05
	7.0	37.03
	8.0	41.91
	9.0	46.69
	10.0	51.36
260 x 180	4.0	26.81
	5.0	33.25
	6.0	39.59
	7.0	45.83
300 x 150	4.0	27.43
	5.0	34.03
	6.0	40.53
	7.0	46.93
	8.0	53.22
	9.0	59.40
	10.0	65.49
300 x 200	4.0	30.57
	5.0	37.96
	6.0	45.24
	7.0	52.42
	8.0	59.50
	9.0	66.47
	10.0	73.34



Steel India Co.
(CHENNAI) PVT. LTD.

WEIGHT & THICKNESS OF S.S GUAGE PIPES

steelindialtd@yahoo.com

DIMENSION		10 SWG (3.2MM)	12 SWG (2.6MM)	14 SWG (2.1MM)	16 SWG (1.65MM)	18 SWG (1.2MM)	19 SWG (1.0MM)
Size	OD	Weight/MTR	Weight/MTR	Weight/MTR	Weight/MTR	Weight/MTR	Weight/MTR
1/2"	12.7	0.754	0.651	0.552	0.452	0.342	0.290
5/8"	15.875	1.006	0.856	0.718	0.582	0.437	0.369
3/4"	19.05	1.258	1.061	0.883	0.712	0.531	0.448
1"	25.4	1.762	1.470	1.214	0.972	0.720	0.605
11/4"	31.75	2.266	1.880	1.544	1.232	0.909	0.763
11/2"	38.1	2.770	2.289	1.875	1.492	1.098	0.920
2"	50.8	3.778	3.108	2.537	2.012	1.476	1.235
21/2"	63.5	4.786	3.928	3.198	2.531	1.854	1.550
3"	76.2	5.794	4.747	3.860	3.051	2.232	1.865
31/2"	88.9	6.802	5.566	4.521	3.571	2.610	2.180
4"	101.6	7.810	6.385	5.183	4.091	2.988	2.495

DIMENSIONAL TOLERANCES OF PIPE ANSI B 36.10 / B 36.19

Tubes Specification ASTM	Nominal Pipe Size (mm)	Permissible variations in Outside Diameter (mm)		Permissible variations in THK	Permissible variations in Length (mm)		Straight tolerance
		Over	Under		Over	Under	Max. curvature in any meter length
A 106 CS Seamless Pipe for High Temp. A 312 Seamless & Welded Austinitic SS Pipes A333 Seamless & Welded pipe for LT Service A 335 Seamless Ferritic Alloy Steel Pipe for high Temp. Service	3 to 40 Incl.	0.4	0.8	-12.5% Except for Welded pipe will filler Metal Added	6	0	Up to 125 mm O/D and 12 mm THK Pipe -0.76 mm Over 125mm O/D to 200 mm O/D Inclusive -1.15 mm Over 200 mm O/D to 324 mm O/D Inclusive -1.52 mm
	Over 40 to 100 Incl.	0.8	0.8				
	Over 100 to 200 Incl.	1.6	0.8				
	Over 200 to 450 Incl.	2.4	0.8				
	Over 450 to 650 Incl.	3.2	0.8				
	Over 650 to 850 Incl.	4.0	0.8				
	Over 850 to 1200 Incl.	4.8	0.8				
A 358 ERW Austinitic Cr-NiAs Pipe for High Temp. serv.	All Sizes (upto 200 NB)	+0.5% (Based on Circum-ferencial Measurement)	-0.5 &%	-0.3 mm			3 mm 3 / meters
A409 ERW Large Dia. Austinitic Steel Pipes	450 to 750 (SCH 5 S & 10 S)	+0.25% +0.4% Based on Circumferential	+0.2 % (For T< 4.8mm) -0.4% (For T 4.8mm) Measurement	-0.46 mm			4.8 mm /3 meters



Formula of Calculating Weight

Weight of Stainless Steel Pipe

O.D. (mm) - Wall Thick (mm) x Wall Thick (mm) x 0.0248 = Wt. per Mtr.
O.D. (mm) - Wall Thick (mm) x Wall Thick (mm) x 0.00756 = Wt. per Feet

Weight of Stainless Steel Round Bar

Dia (mm) x Dia (mm) x 0.00623 = Wt. per Mtr.
Dia (mm) x Dia (mm) x 0.0019 = Wt. per Feet

Weight of Stainless Steel Square Bar

Dia (mm) x Dia (mm) x 0.00788 = Wt. per Mtr
Dia (mm) x Dia (mm) x 0.0024 = Wt. per Feet

Weight of Stainless Steel hexagonal Bar

A/F (mm) x A/F (mm) x 0.00680 = Wt. per Mtr.
A/F (mm) x A/F (mm) x 0.002072 = Wt. per Feet

Weight of Stainless Steel Flat Bar

Width (mm) x Thickness (mm) x 0.00798 = Wt. per Mtr.
Width (mm) x Thickness (mm) x 0.00243 = Wt. per Feet

Weight of stainless Steel Sheets & Plates

Length (Mtrs) x Width (Mtrs) x Thick (mm) x 8 = Wt. per PC
Length (Feet) x Width (Feet) x Thick (mm) x 3/4 = Wt. per PC

Weight of Stainless Steel Circle

Dia (mm) x Dia (mm) x Thick (mm) 160 = Gms. per PC
Dia (mm) x Dia (mm) x Thick (mm) 0.0000063 = kg. per PC

Weight of Brass Metal

Weight of Stainless Steel + 9%

Weight of Copper Metal

Weight of stainless Steel + 12%

Weight of Aluminum Metal

Weight of stainless Steel ÷ 3

Conversion

1 Metre = 3.2808 Foot
1 Feet = 304.8 mm
1 Inch = 25.4 mm

Barlow's Formula for calculating bursting pressure

P=2ST/D or t-DP/2S or S-DP/2T or D=2ST/P
P= Bursting pressure Psi
S = Tensile Strenght of tube
T = Wall thickness (in inches)
D = Outside Diameter (in inches)

*** Formula for healthy Business**

Honesty + Quality Goods + Quick service + Reasonable Rate = Good Healthy Business

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APPLICATION INDUSTRIES







- PETRO CHEMICAL PLANTS
- ACID & CHEMICAL INDUSTRIES
- AUTOMOBILE INDUSTRIES
- HEAVY MACHINERIES
- BEVERAGES & DISTILLERIES
- CEMENT PLANTS
- ELECTRIC &ELECTRONIC INDUSTRIES
- FERTILIZER INDUSTRIES
- FOOD & DAIRY INDUSTRIES
- OLD & GAS INDUSTRIES
- PAPER & PULP INDUSTRIES
- PHARMACEUTICALS
- POWER SECTORS
- REFINERIES
- SUGAR MILLS

STEEL INDIA

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